Darren Biskup

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EDUCATION

Columbia University — New York, NY
M.S./Ph.D. Mechanical Engineering (Robotics & Rehabilitation) — Aug 2024 - Present
Columbia-Dream Sports AI Ph.D. Fellowship — GPA: 4.00/4.33

University of Illinois, Urbana-Champaign — Urbana, IL
B.S. Mechanical Engineering, Computer Science Minor — May 2024
High Honors, James Scholar Honors Program, Dean's List — GPA: 3.95

SKILLS

Languages & Software: C/C#, Python (PyTorch, OpenCV), CATIA 3DEXPERIENCE, SolidWorks, ROS2, Git
Controls & Estimation: Model Predictive Control, Trajectory Optimization, Sensor Fusion (EKF/UKF), MuJoCo
Hardware & Design: CAN/CAN-FD BLDC Control, Design for Injection Molding, GD&T

EXPERIENCE

Robotics And Rehabilitation (ROAR) Lab, Columbia University — Aug 2024 - Present
Graduate Research Assistant

- Engineered the hierarchical control architecture for a cable-driven physical assistance robot, integrating VICON motion capture and force plate sensing. Closes loop on desired net body wrench by synchronizing multi-axis motor commands via CAN bus with inline load cell tension feedback.
- Authored IEEE RA-L submission detailing a real-time, Ground Reaction Force-aware convex MPC leveraging human centroidal dynamics for underactuated upright assistance. Validated against a fully-actuated impedance baseline, the architecture reduced dynamic tracking error by 24% and mechanical work by 29%.
- Designing force guidance protocols for dynamic yoga pose training based on comparative analysis of experimentally collected expert-novice movement data, leveraging automated kinematic assessment.

Mehta Research Group/GazzolaLab, UIUC — Jan 2023 - May 2024
Undergraduate Research Assistant - Soft Robotics

- Built a joystick-controlled robotic soft arm with parallel fiber-reinforced actuators, using a Raspberry Pi and GameCube controller to drive pneumatic solenoid valves.
- Designed and built a 3D calibration grid using laser-cut acrylic and green paint; developed Python code utilizing OpenCV and scikit-image to detect colored blobs and extract marker coordinates.
- Developed scripts to input arm pose maneuver instructions to allow rigorous comparison between simulation and real-world.

Skydio — May 2022 - Aug 2022
Product Design Engineering Intern

- Design for Injection Molding: designed the Navigation Camera mounting mechanism for the company's next generation performance quadcopter drone. Communicated with overseas vendors in China to implement design change, ensuring injection moldability and minimal lead time.
- Designed and prototyped the mobile tablet adapter for the next generation drone controller using FDM and SLS 3D-Printing. Received DFM feedback from injection molding vendor.
- Worked with Manufacturing Engineers to design wire routing layout for drone RF cables, board to board connections, 3-phase motor power cables, LED cables.

Lucid Motors — May 2021 - Aug 2021
Mechanical Engineering Intern, High Voltage

- Battery Pack Development: Improved design for the high voltage chain halving the number of bolts required to join busbars. Conducted thermal analysis on busbar joints with new design to evaluate the new design's effect on car's horsepower, range, efficiency, and thermal endurance.
- Collected and analyzed data on heat generation from bolted busbars joints, and used this data to calculate theoretical horsepower and efficiency loss of specific joints.

PROJECTS

Senior Capstone Project - Neurosurgical Balloon Tunnelling Device

- Developed an innovative shunt passage method using expandable balloons for ventriculoperitoneal operations which minimizes tissue damage and backtracking leading to improved patient and surgeon comfort.